NeurIPS 2026 Submission Structure and Checklist Report

Deep strategy guide for the LLM emotion geometry -> brain representation comparison paper. This is not a draft manuscript; it is a submission-control document explaining exactly how to structure the paper, where each result belongs, what each section must prove, and how to satisfy NeurIPS review and checklist expectations.

Central paper thesis: uncertainty in emotion representations is better explained by relative geometric competition between class prototypes than by absolute distance from a centroid. LLM experiments establish the geometry and logit coupling; the five-emotion brain analysis tests whether the same geometric logic is observable in neural representations.

Item	Recommended framing
Contribution type	General submission with a strong conceptual/mechanistic contribution. Avoid submitting as pure Theory, Use-Inspired, or Negative Results. Concept & Feasibility language can be used cautiously, but General is safer because the evidence is empirical and complete.
What is new	A systematic geometry-to-decision account: LLM emotion embedding geometry is not only clustered; overlap regions align with logits, and relative centroid margin predicts uncertainty. Brain data provides biological validation of the same geometric competition idea.
What not to claim	Do not claim LLMs replicate the brain; do not claim universality across all emotions or models; do not claim distance alone equals ambiguity; do not present VA alignment as the main proof.
Submission target	A principle-first empirical paper: problem -> failed intuition -> margin principle -> LLM proof -> brain validation -> implications and limitations.

1. Executive strategy: what the paper must convince reviewers of

The strongest version of this paper is not a descriptive comparison of LLMs and brain data. It should be written as a mechanism-discovery paper. Reviewers should finish the paper remembering one thing: representation geometry has decision relevance because uncertainty emerges from centroid competition. The LLM phases show how emotion structure evolves through training, depth, and dimensional compression. Phase 5 closes the mechanistic loop by showing that geometric overlap corresponds to logit behaviour. The brain analysis then asks whether this geometric principle is also visible in biological emotion representations.

This framing directly targets the NeurIPS criteria. Quality is supported by formal metrics, controls, and statistical testing. Clarity is supported by explicit definitions and a reproducible pipeline. Significance comes from turning embedding geometry into a reusable explanation of uncertainty rather than a visualization. Originality comes from connecting centroid geometry, logit behaviour, dimensional compression, and brain representation structure in one coherent framework.

Reviewer-facing one-sentence contribution: We show that ambiguity in emotion representations is governed by relative centroid competition rather than absolute distance, and that this geometric competition predicts both LLM logits and classifier uncertainty in brain-derived representations.

2. Exact narrative arc recommended for the paper

Narrative step	Purpose	Evidence/result to use	What reviewer should think
1. Representations are geometric	Ground the paper in a known but important premise: embeddings and neural patterns form high-dimensional spaces where distances and directions matter.	Related work on embeddings, RSA, brain/LLM alignment; Phase 1 silhouette; brain RDM/centroids.	This is a legitimate representation-geometry problem, not a visualization exercise.
2. Simple geometry is insufficient	Show intellectual honesty by testing the naive assumption that distance from class centroid equals uncertainty.	Brain: distance vs uncertainty Spearman r=0.1509, AUC=0.5627. Density p≈0.46 as supporting negative result.	The authors are not assuming their metric; they falsify the obvious alternative.
3. Introduce centroid competition	Define margin as relative distance to the correct centroid versus nearest competing centroid.	$A_i = d_{correct} - d_{nearest\_other}$; or use sign-reversed G_i consistently.	Ambiguity is a relational property, not a radial property.

Narrative step	Purpose	Evidence/result to use	What reviewer should think
4. Prove it in LLMs	Use LLMs as the cleaner controllable system: layers, training state, logits, and dimensional interventions are available.	Phase 1-5: silhouette, overlap, signal retention, 20D subspace, logit consistency.	The phenomenon is systematic across architectures and training states.
5. Validate with brain data	Use brain data to test whether a related geometric principle appears in biological emotion representations.	LOSO 0.56 vs chance 0.20; margin r=0.6108, p=7.78e-22; AUC=0.8124; shuffle r=0.0287.	The brain analysis strengthens generality without overclaiming brain equivalence.
6. Explain differences	Show that shared local geometry can coexist with different global organization.	VA: brain PC1-arousal r=0.965, p=0.0078; RDM/centroid differences.	The authors understand both convergence and divergence between systems.
7. Conclude with reusable principle	Present the result as a metric/framework others can use.	Margin, overlap, logit consistency, low-rank subspace tests.	This is useful beyond this one dataset.

3. Recommended paper structure in substantial detail

3.1 Title and abstract strategy

The title should foreground the mechanism, not the dataset. Titles centred on "LLM-brain comparison" will sound descriptive; titles centred on "geometric competition" sound like a contribution. The abstract must include the negative result (distance alone fails), the new principle (margin), the LLM mechanistic validation (logits), and the brain validation.

Component	What it must do	Recommended content
Title	Signal the mechanism clearly	"Geometric Competition Predicts Ambiguity in Language Model and Neural Emotion Representations" or "Centroid Margin as a Mechanism for Ambiguity in Artificial and Neural Representations".
Abstract sentence 1	State the broad problem	Representations in LLMs and brains are often studied geometrically, but it remains unclear how uncertainty emerges from that geometry.
Abstract sentence 2	State failed intuition	We show that absolute distance to a class prototype is a weak predictor of uncertainty.
Abstract sentence 3	State new metric	We propose relative centroid margin as a measure of representational competition.
Abstract sentence 4	State LLM proof	Across BGE and MPNet variants, fine-tuning sharpens cluster geometry, isolates emotion into a compact subspace, and overlap regions align strongly with logits.
Abstract sentence 5	State brain validation	In five-emotion fMRI ROI representations, margin strongly predicts classifier uncertainty and collapses under feature shuffle.
Abstract closing	State impact	These findings suggest a local geometric principle for ambiguity that generalizes across artificial and neural representations while allowing differences in global organization.

3.2 Introduction structure

Paragraph	Goal	Must include	Avoid
P1: Motivation	Explain why representation geometry matters to ML and cognitive neuroscience.	Embedding spaces, semantic structure, brain RSA, need to interpret internal representations.	Do not open with too many implementation details.
P2: Gap	Identify missing mechanism.	Prior work shows geometry/alignment, but not how ambiguity/uncertainty arises from geometry.	Do not say no one has ever studied uncertainty; be precise.
P3: Failed intuition	Introduce the naive distance hypothesis.	The tempting hypothesis: farther from centroid -> more uncertain. Then say this is incomplete.	Do not bury this; it is a key contribution.
P4: Proposed principle	Introduce margin competition.	Ambiguity depends on distance to nearest competitor relative to true centroid.	Do not define multiple competing metrics here; keep it clean.
P5: Contributions	List 3-4 explicit contributions.	(1) LLM geometry evolution; (2) low-rank affective subspace; (3) geometry-logit coupling; (4) brain validation of margin.	Do not include every minor analysis as a contribution.

3.3 Related work structure

Related work should be organized around the logic of the paper, not around chronology. The goal is to establish that prior work justifies the problem but leaves your precise mechanism open.

Subsection	Purpose	Examples of literature to cite	How it sets up your novelty
LLM representation geometry	Show that embeddings encode meaning in geometric spaces.	Word embeddings, contextual embedding anisotropy, probing, mechanistic interpretability, semantic directions.	Prior work studies structure; your work tests ambiguity and logits through centroid competition.
Neural representational geometry	Ground brain part in accepted neuroscience methodology.	RSA, MVPA, fMRI pattern analysis, emotion representation studies, NeuroEmo / visually evoked emotion dataset.	Prior work compares representational structure; your work validates a margin-uncertainty relation.
Brain-LLM alignment	Position cross-system comparison cautiously.	Brain predictivity of language models, RSA between DNNs and brain activity, scaling laws for brain alignment.	Prior work asks whether representations align; your paper asks whether a local geometric mechanism is shared.
Uncertainty and margins	Justify why margin is theoretically meaningful.	Decision boundaries, logistic regression margins, probabilistic classification, calibrated uncertainty.	Your result links prototype competition with empirical uncertainty in representations.

3.4 Problem formulation and definitions

This section should appear before Methods or as the first part of Methods. It is essential for NeurIPS clarity because reviewers need to know exactly what x, c, d, margin, uncertainty, overlap, and RDM mean across both LLM and brain spaces.

Representation vector: $x_i \in R^d$. For LLMs, $d=768$ in the full embedding space and $d=20$ in the isolated affective subspace. For brain, $d=48$ ROI features.

Centroid: $c_k = (1/N_k) \sum_{\{i:y_i=k\}} x_i$.

Euclidean distance: $d(x_i,c_k) = ||x_i - c_k||_2 = \mathrm{sqrt}(\sum_{\{j=1\}}^d (x_{ij} - c_{kj})^2)$.

Ambiguity margin convention for the final paper: $A_i = d(x_i,c_{\{y_i\}}) - \min_{\{k\neq y_i\}} d(x_i,c_k)$. Larger A_i means the sample is less clearly closer to its true centroid; values near zero mark boundary competition; positive values mean a competing centroid is closer.

Classifier uncertainty: $u_i = 1 - P(y_i|x_i)$. Use this when a classifier gives probabilities, especially in the brain validation where the dataset itself does not contain confidence scores.

Logit margin for LLMs: $L_i = z_{true} - z_{other}$. If using the sign-reversed geometric separation margin $G_i = d_{other} - d_{true}$, then positive G_i and positive L_i both mean the true class is favoured. State the convention explicitly.

3.5 Methods section structure

Methods subsection	Required content	Specific details from this project
LLM dataset and embeddings	Dataset, labels, balancing, models, layers, pooling, normalization.	4038 validation samples; labels anger/fear/happiness/love/sadness/surprise; BGE and MPNet; base vs fine-tuned; final layer 12 and middle layer 6; mean pooling; 768D embeddings; L2 normalization where used.
LLM geometry pipeline	Centroids, distances, silhouette, overlap, radial bins, signal retention, subspace extraction.	Phases 1-4 should be described as a coherent pipeline: topology -> overlap -> direction removal -> 20D affective subspace.
LLM logit consistency	How logits are compared with geometric overlap.	Overlapping points: $d_{wrong} < d_{true}$; logit agreement: $z_{wrong} > z_{true}$; distance-logit correlation: $corr(G_i,L_i)$.
Brain data pipeline	Dataset source, subjects, preprocessing, feature construction.	NeuroEmo/OpenNeuro ds005700 v1.2.0; 40 subjects; five emotions; 200 subject-emotion rows; 48 Harvard-Oxford cortical ROI features; BOLD/ROI extraction; scaled features.
Brain validation	Decoding and margin validation.	LOSO decoding, bootstrap CI, permutation test, centroid/RDM, margin vs uncertainty, feature-shuffle control, VA PCA.
Statistics	Exact tests and why used.	Spearman/Pearson correlations, AUC, permutation p-values, bootstrap CIs, silhouette, accuracy, balanced accuracy.
Reproducibility	Hardware, seeds, packages, code release.	MacBook Pro M4 Pro, 24GB, macOS 15.3.1; CPU for brain; LLM compute to fill; anonymized code/data package.

4. Results structure: what each result section must prove

4.1 Results section 1 - LLM geometry emerges and sharpens with supervision

This section uses Phase 1 and part of Phase 2. The goal is to establish that the LLM spaces contain structured emotion geometry, and that training state/layer depth affects that geometry. This is the foundation; do not yet make the strongest ambiguity claim here.

Evidence	Exact outputs	How to interpret
Silhouette ranking	MPNet-FT-Final 0.8597; BGE-FT-Final 0.6847; MPNet-FT-Mid 0.1434; BGE-Base-Final 0.0570; BGE-FT-Mid 0.0486; MPNet-Base-Final 0.0471; MPNet-Base-Mid 0.0366; BGE-Base-Mid 0.0201.	Fine-tuned final layers form highly separated affective islands; base/mid-layer spaces are more cloud-like.
Training effect	Fine-tuned final layers dominate all base and mid-layer variants.	Supervision acts as a geometric compressor that sharpens class boundaries.
Architecture effect	MPNet generally shows higher topological separation than comparable BGE variants.	Architecture contributes to baseline geometry, but training state is the main effect.

4.2 Results section 2 - Overlap and radial structure reveal ambiguity zones

This section uses Phase 2 but should be written carefully. Phase 2 originally stated that distance is a near-perfect proxy for uncertainty. Your later brain validation showed this is too strong. The correct paper framing is: radial distance reveals overlap zones, but relative margin is the validated ambiguity mechanism.

Evidence	Exact outputs	Paper-safe interpretation
Global overlap	BGE-Base-Final 19.22%; MPNet-Base-Final 18.45%; BGE-FT-Final 1.81%; MPNet-FT-Final 0.42%; BGE-Base-Mid 28.14%; MPNet-Base-Mid 16.92%.	Fine-tuning greatly reduces geometric overlap; pretrained spaces contain more boundary entanglement.
Three radial zones	Core r<0.6 has 0% overlap; transition 0.61.5 has overlap >30-50%.	Embedding space has structured core-to-boundary organisation.
Pairwise confusions	Happiness-Love 42% overlap in pretrained models; Fear-Anger 12%.	Some emotion pairs are intrinsically more entangled in pretrained semantic space.
Certainty buffer	MPNet-FT-Final +1.875; BGE-FT-Final +1.750; MPNet-Base-Final 0.000; BGE-Base-Final +0.125.	Fine-tuning creates a low-overlap buffer between density peak and ambiguity onset.

4.3 Results section 3 - Emotional information is low-rank but training changes redundancy

This section uses Phase 3. Its function is to show that emotion is not simply spread uniformly across the embedding; it is concentrated in discriminative linear directions. This supports a stronger mechanistic story because if removing a small number of directions erases classification, the geometry is organized around specific axes.

Evidence	Exact outputs	Interpretation
Baseline accuracy	MPNet-FT-Final 98.14%; BGE-FT-Final 98.14%; MPNet-Base-Final 95.17%; BGE-Base-Final 95.17%.	Emotion classes are highly linearly decodable in final-layer embeddings.
After removing 5 dims	MPNet-FT 93.69%; BGE-FT 94.55%; MPNet-Base 76.24%; BGE-Base 76.86%.	Base models lose more early signal; fine-tuned models retain accuracy initially.
After removing 10 dims	MPNet-FT 51.98%; BGE-FT 63.24%; MPNet-Base 52.85%; BGE-Base 40.22%.	Major accuracy collapse begins after top directions are removed.
After removing 15 dims	MPNet-FT 17.70%; BGE-FT 24.75%; MPNet-Base 34.41%; BGE-Base 26.49%.	MPNet-FT nearly reaches chance by 15 removed directions.
After removing 20 dims	MPNet-FT 16.58%; BGE-FT 16.71%; MPNet-Base 26.61%; BGE-Base 22.15%.	Fine-tuned final-layer signal is effectively erased by 15-20 directions.
Erasure point	MPNet-FT dim 15; BGE-FT dim 20; MPNet-Base dim 67; BGE-Base dim 26.	Fine-tuning compresses the affective signal; pretrained MPNet distributes it more redundantly.

4.4 Results section 4 - A compact 20D affective subspace preserves the signal

This section uses Phase 4. It is one of the most important LLM results because it shows that the geometry is not an artifact of high-dimensional visualization. Emotion structure becomes clearer when projected into the subspace identified by classifier-weight SVD.

Evidence	Exact outputs	Interpretation
Silhouette improvement	MPNet-FT 0.8597 -> 0.9068 (+5.5%); BGE-FT 0.6847 -> 0.8353 (+22.0%); MPNet-Base 0.0471 -> 0.4075 (+765.2%); BGE-Base 0.0570 -> 0.4146 (+627.4%).	The affective subspace denoises the apparent high-dimensional cloud, especially in base models.
Accuracy retention	MPNet-FT 98.24% -> 98.27%; BGE-FT 97.99% -> 98.09%; MPNet-Base 94.08% -> 95.64%; BGE-Base 93.78% -> 95.12%.	20D retains essentially all emotion-classification signal and can slightly improve base-model probing.
Relational logic	Closest pairs: Happiness-Love and Fear-Anger. Furthest pairs: Sadness-Surprise and Happiness-Fear.	The low-rank subspace preserves meaningful affective relationships.

4.5 Results section 5 - Geometry is coupled to logits

This should be the strongest LLM result. It connects embedding geometry to model decisions and prevents the reviewer from saying the geometry is just post-hoc visualization. Use the exact Phase 5 outputs.

Metric	BGE-FT-Final	MPNet-FT-Final	Interpretation
Total validation samples	4038	4038	Same evaluation size for both fine-tuned final-layer models.
Overlap count	76	71	Only a small number of samples lie closer to an incorrect centroid after fine-tuning.
Overlap percent	1.882%	1.758%	Fine-tuned overlap is rare but informative.
Logit agreement rate	0.9342	1.0000	When geometry says a point is closer to the wrong centroid, logits almost always agree.
Average logit difference in overlap	-3.3245	-3.4174	The wrong/competing class is not weakly favoured; the model is confidently wrong in overlap regions.
Distance-logit correlation	0.9572	0.9884	Extremely strong alignment between geometric margin and logit margin.
Correlation p-value	0.0 output; report as p < 1/(B+1) or p < 0.001 depending on implementation	0.0 output; report as p < threshold	Do not report p=0 literally; use a lower-bound convention.

How to write this result: "Geometric overlap was not merely a spatial artifact: in fine-tuned final-layer models, samples closer to an incorrect centroid were assigned logits favouring that competing class in 93.4% (BGE) and 100% (MPNet) of overlap cases. Moreover, geometric margin and logit margin were strongly correlated (r=0.957 and r=0.988), indicating tight coupling between embedding-space geometry and decision-layer behaviour."

4.6 Results section 6 - Brain validation of the geometry principle

The brain section must not be presented as an afterthought. It provides biological validation, but with careful scope. It should show: (1) the brain representation contains emotion signal, (2) raw distance alone is weak, (3) margin predicts uncertainty, (4) shuffling destroys the relationship, and (5) global brain axes can differ from LLM axes.

Brain result	Exact output	Interpretation
Dataset representation	200 subject-emotion observations; 40 subjects x 5 emotions; 48 ROI features.	Five-emotion brain representation space for Afraid, Calm, Delighted, Depressed, Excited.
LOSO decoding	Accuracy 0.5600; balanced accuracy 0.5600; chance 0.2000.	Brain ROI patterns contain emotion information that generalizes to held-out subjects.
Bootstrap CIs	Accuracy CI [0.4900, 0.6300]; balanced accuracy CI [0.4906, 0.6280].	Decoding performance is robustly above chance.
Permutation null	Mean 0.197825; SD 0.0334364; p output 0.0, report p<0.001 if 1000 permutations.	Emotion signal is not due to random label structure.
Raw distance vs uncertainty	Spearman r=0.1509, p=3.29e-02; Pearson r=0.1471, p=3.76e-02; AUC=0.5627.	Absolute distance is weak; this explicitly rejects the naive radial ambiguity account.
Margin vs uncertainty	Spearman r=0.6107627691, p=7.7796e-22; Pearson r=0.6351; AUC=0.8124.	Relative centroid competition strongly predicts uncertainty.
Feature shuffle	Shuffled r=0.0287, p=0.687.	The margin-uncertainty relationship depends on real multivariate structure.
VA axis	PC1 arousal r=0.965, p=0.0078; PC1 valence r=0.255, p=0.6789.	Dominant brain centroid axis is arousal-aligned; use as global-structure support, not main proof.
Density p-value	p≈0.46.	Global density alone does not explain ambiguity; supports the move from density/distance to margin.

5. Figure strategy for a 9-page NeurIPS paper

A strong NeurIPS paper should not include every plot. It should include the plots that make the mechanism unavoidable. Use supplementary material for exhaustive per-model phase plots.

Main figure	Panels	Purpose	Where it goes
Figure 1: Concept and pipeline	A: centroid/distance/margin schematic; B: LLM pipeline phases; C: brain validation pipeline.	Lets reviewers understand the metric and the cross-system design immediately.	End of Introduction or start of Methods.
Figure 2: LLM geometry evolves through training	A: silhouette ranking; B: overlap percentages; C: certainty buffer; D: PCA/UMAP illustrative plot.	Shows training and layer effects on geometry.	Results 1.
Figure 3: Low-rank affective core	A: accuracy decay curve; B: erasure points; C: 20D silhouette improvements; D: 20D accuracy retention.	Shows emotional signal is concentrated and recoverable in compact subspace.	Results 2.
Figure 4: Geometry predicts logits	A: overlap definition; B: logit agreement; C: distance-logit correlation; D: example overlap samples.	Strongest LLM mechanism proof.	Results 3.
Figure 5: Brain validation	A: LOSO decoding vs permutation null; B: distance vs uncertainty weak; C: margin vs uncertainty strong; D: feature shuffle collapse.	Shows biological validation and addresses the distance-assumption critique.	Results 4.

Main figure	Panels	Purpose	Where it goes
Figure 6: Global structure comparison	A: brain VA alignment; B: RDM heatmap; C: LLM/brain conceptual comparison.	Explains convergence/divergence. Optional in main text if space permits.	Discussion or appendix.

6. NeurIPS checklist: exact planning guide

Checklist item	Recommended answer	What the paper must include
Claims	Yes	Abstract and Introduction must state only supported claims: geometry-logit coupling in tested LLMs; margin-uncertainty relationship in five-emotion brain data; no universal brain equivalence claim.
Limitations	Yes	Separate Limitations section. Mention emotion category count, dataset specificity, centroid assumption, observational nature, VA n=5 centroid issue, and need for broader brain/LLM datasets.
Theory/proofs	N/A unless formal theorem included	If you include equations only as definitions, answer N/A. If you state any proposition about margins, include assumptions and proof sketch in appendix.
Experimental reproducibility	Yes	Provide code, exact dataset links, model names, layers, seeds, pooling, splits, and scripts for figures.
Open access data/code	Prefer Yes	Anonymized supplementary zip or anonymous repository. Public data links for OpenNeuro and model sources. README with commands.
Experimental details	Yes	LLM: BGE/MPNet variants, layer 6/12, base/FT, mean pooling, 4038 validation samples, 6 labels. Brain: 40 subjects, 5 emotions, 48 ROIs.
Statistical significance	Yes	Report correlations, AUCs, permutation p-values, bootstrap CIs, shuffle controls. State what each CI/error bar captures.
Compute resources	Yes	MacBook Pro M4 Pro, 24GB, macOS 15.3.1 for brain CPU workflow; add LLM compute details from collaborator. Runtime table.
Code of Ethics	Yes	Public de-identified brain data; no new human subjects; no deployment; no high-risk model release.
Broader impacts	Yes or N/A with explanation	Foundational interpretability work. Discuss risk of overinterpreting brain-LLM similarity and avoid biological claims unsupported by evidence.
Safeguards	N/A unless releasing model	If only releasing analysis code/derived embeddings, state no high-risk model is released.
Licenses	Yes	Cite OpenNeuro dataset, NeuroEmo paper, model checkpoints and licenses, code libraries. State versions.
Assets	Yes if releasing processed matrices/code	Document processed embeddings/brain matrices, provenance, limitations, and license. Otherwise N/A if no new assets are released.
Crowdsourcing/human subjects	N/A for new collection	No crowdsourcing. No new human subjects. Original dataset authors handled recruitment/consent.
IRB approvals	N/A for your team unless institution requires secondary review	State no new human-participant data collection; public de-identified OpenNeuro data only.
LLM usage declaration	Yes for core method	LLMs are analysed objects. Separately, writing/editing assistance does not need declaration unless required by final policy.

7. Reviewer attack surface and how to pre-empt it

Likely reviewer criticism	Why it could hurt	Pre-emptive fix in paper
"This is just visualization."	Could lower significance/originality.	Make Phase 5 central: geometry predicts logits with r=0.957/0.988 and high agreement.
"Distance does not imply ambiguity."	This is a serious methodological critique.	Agree and show you tested it: raw distance weak; margin strong. Make this a contribution.
"Correlation is not causation."	Could weaken mechanism language.	Use "coupled", "aligned", "predictive" unless you perform interventions. Signal-removal and Phase 5 strengthen but do not prove causality.
"Brain dataset is small / only five emotions."	Could limit generalization.	Frame brain as biological validation, not complete neuroscience proof. Include limitations.
"VA analysis has only five points."	Could be overinterpreted.	Use VA as support for global organization only; state n=5 centroid caveat.
"What is novel compared with RSA/probing?"	Novelty risk.	State novelty is not geometry itself but margin-based ambiguity + logit coupling + brain validation.
"Why Euclidean distance?"	Metric choice challenge.	Justify L2 normalization and Euclidean distance; include cosine sensitivity in appendix if possible.
"Fine-tuned models obviously cluster."	Risk of triviality.	Emphasize low-rank compression, overlap-logit consistency, and margin competition; clustering is only the first step.

8. Exact action checklist before submission

Priority	Task	Owner / status	Why it matters
Critical	Lock one margin sign convention across LLM and brain.	Paper team	Prevents confusion in formulas and interpretation.
Critical	Run/directly confirm LLM margin-logit stats for both BGE and MPNet in final table.	LLM side	Phase 5 is the mechanistic anchor.
Critical	Prepare Figure 4 geometry-logit panel.	Figure owner	This is likely the strongest acceptance-driving figure.
Critical	Write Methods with enough detail to reproduce.	Paper team	Directly affects NeurIPS Clarity and Quality scores.
Critical	Create anonymized code/data supplement.	Code owner	Reviewers may penalize weak reproducibility.
High	Add CIs or shaded bands to central plots where possible.	Stats owner	Satisfies statistical significance checklist.
High	Add limitations section before reviewers ask.	Writing owner	Honesty is rewarded and protects claims.
High	Check licenses for OpenNeuro, BGE, MPNet, datasets.	Submission owner	Checklist item 12.
High	Add compute reporting table.	Submission owner	Checklist item 8.
Medium	Sensitivity check with cosine distance or normalized embeddings if feasible.	Stats owner	Pre-empts metric-choice critique.
Medium	Move exhaustive plots to appendix.	Figure owner	Protects main paper clarity.
Medium	Create one-page glossary for BOLD, ROI, RDM, margin, logits.	Appendix owner	Improves accessibility for ML reviewers unfamiliar with fMRI.

9. Recommended final paper page budget

Page allocation	Content
Page 1	Title, abstract, intro opening, contributions.
Page 2	Related work and problem formulation.
Page 3	Methods overview and Figure 1 pipeline.
Page 4	LLM topology and overlap results (Phases 1-2).
Page 5	Low-rank subspace and signal retention (Phases 3-4).
Page 6	Geometry-logit consistency (Phase 5). This page should be extremely strong.
Page 7	Brain validation: decoding, margin vs uncertainty, shuffle control.
Page 8	Global structure: RDM/VA, synthesis across LLM and brain.
Page 9	Discussion, limitations, conclusion.
Appendix	Full formulas, statistical tables, all phase reports, extra plots, glossary, compute, ethics/checklist support.

10. Final interpretation hierarchy

The paper needs a hierarchy of claims so the strongest statements are protected by the strongest evidence. Use this hierarchy consistently:

Level	Claim strength	Supported statement	Evidence
Primary	Strong	In LLMs, emotion geometry is coupled to logits and decision behaviour.	Phase 5: agreement 0.934/1.000; correlation 0.957/0.988.
Primary	Strong	In brain data, margin predicts classifier uncertainty better than distance.	Margin r=0.6108, AUC=0.8124; distance AUC=0.5627; shuffle r=0.0287.
Secondary	Strong	Fine-tuning sharpens and compresses LLM emotion geometry.	Silhouette, overlap, erasure points, 20D results.
Secondary	Moderate	LLM emotion signal is low-rank.	20D retains ~98% accuracy; removal of top directions erases signal.
Supporting	Moderate	Brain global geometry appears arousal-aligned.	PC1-arousal r=0.965, p=0.0078, but n=5 centroids.
Supporting	Cautious	Density alone is insufficient.	Density p≈0.46; use as negative/supporting result.

11. Detailed section-by-section build plan

This section gives a build plan for the actual manuscript. It is not prose to paste directly; it is an instruction map for what each paragraph, table, and figure must accomplish. Use it as a writing checklist when drafting the paper.

Paper section	Paragraph / component	Precise content to include	Acceptance purpose
Abstract	Sentence 1	Representational geometry is increasingly used to interpret LLMs and neural data, but the relationship between geometry and uncertainty remains under-specified.	Signals broad relevance.
Abstract	Sentence 2	We test whether ambiguity is explained by absolute distance from emotion centroids or by relative competition with the nearest alternative centroid.	Frames a falsifiable problem.
Abstract	Sentence 3	Across BGE and MPNet variants, fine-tuning sharpens emotion clusters, reduces overlap, and isolates a compact affective subspace.	Summarizes LLM evidence.
Abstract	Sentence 4	Overlap regions align strongly with logits, showing that embedding geometry predicts model decisions.	Mechanistic contribution.
Abstract	Sentence 5	In brain ROI data, margin strongly predicts uncertainty and survives cross-validated validation while feature shuffle removes the effect.	Cross-system validation.
Introduction	Opening	Start with the problem of understanding internal representations, not with the dataset.	Makes paper relevant to NeurIPS broadly.
Introduction	Gap	Prior geometry and brain-alignment work identifies structure, but often stops short of explaining decision uncertainty.	Clear novelty gap.
Introduction	Challenge	A simple radial view - farther from centroid means more uncertain - is intuitive but incomplete.	Sets up negative result.
Introduction	Contribution list	List exactly 4 contributions: LLM geometry evolution, low-rank affective core, geometry-logit coupling, brain margin validation.	Reviewer-friendly clarity.
Methods	Problem setup	Define x_i, y_i, K, d, centroid, distance, margin, uncertainty, overlap, RDM.	Reproducibility and mathematical clarity.
Methods	LLM pipeline	Describe all eight variants: BGE/MPNet, base/FT, layer 6/layer 12, 4038 samples, 6 labels, mean pooling.	Prevents vague experimental setting.
Methods	Brain pipeline	Describe NeuroEmo/OpenNeuro, 40 subjects, five emotions, 48 ROI features, subject-emotion averaging, scaling.	Human-data clarity.
Methods	Stats	State exact tests: Spearman, Pearson, AUC, permutation, bootstrap, feature shuffle.	Checklist item 7.
Results	LLM geometry	Report silhouette and overlap before dimensional analyses.	Establishes phenomenon.
Results	Subspace	Report erasure and 20D preservation.	Shows low-rank mechanism.
Results	Logits	Report Phase 5 after geometry/subspace, not before.	Mechanistic climax.
Results	Brain	Report brain validation after LLM mechanism.	Generalization rather than exploratory add-on.
Discussion	Synthesis	Explain local convergence despite global divergence: margin/competition can be shared even if VA axes differ.	Nuanced interpretation.
Limitations	Scope	Emotions, datasets, models, centroid assumptions, observational nature.	Reviewer trust.

12. Main-text wording guardrails

This project is strong, but it can be weakened by overclaiming. The following wording rules should be enforced during drafting.

Avoid saying	Say instead	Reason
LLMs replicate the brain	LLMs and brain representations share a local geometric ambiguity principle under the tested conditions.	Avoids unsupported biological equivalence.
Distance from centroid measures ambiguity	Raw distance is insufficient; relative centroid margin better captures ambiguity.	Your own statistics show this.
The geometry causes logits	Embedding geometry is tightly coupled to logits / predicts logits / aligns with decisions.	Unless you run causal interventions, avoid causal overstatement.
Universal law of emotion geometry	Consistent geometric principle observed across tested LLM and brain representations.	Avoids overgeneralization.
The brain encodes emotion like an LLM	Both systems can be analysed with a shared representational geometry framework.	Safer and more accurate.
Pure emotion centroid	Class prototype / average emotion representation.	Avoids philosophical/psychological overclaim.
p=0.0	$p<0.001$ or $p\leq 1/(B+1)$, depending on number of permutations.	p-values should not be reported as exactly zero.
Density proves ambiguity	Density describes global packing; margin validates local ambiguity.	Prevents density $p=0.46$ issue.

13. Detailed NeurIPS checklist answer plan

The NeurIPS checklist is visible to reviewers. It should not be treated as an administrative afterthought. It is a second route through which reviewers assess rigor and transparency. Below are recommended answers and the exact evidence the paper needs to point to.

Checklist question	Answer	Where to point in paper	Exact supporting content
1. Claims	Yes	Abstract, Introduction, Limitations	Claims limited to tested LLMs and one five-emotion brain dataset; state margin and logit coupling as observed evidence, not universal proof.
2. Limitations	Yes	Dedicated Limitations section	Limited emotion labels; public brain dataset; centroid/prototype assumption; no causal intervention; VA based on five centroids; LLM model family coverage.
3. Theory/proofs	N/A	Problem formulation / appendix	No theorems unless added. Equations are metric definitions. If a proposition is included, add assumptions and proof sketch.
4. Reproducibility	Yes	Methods, Appendix, Supplement README	Dataset links, model names, pooling method, layers, seeds, exact scripts, outputs.
5. Code/data	Yes if suppl ement included	Supplementary material	Anonymized code zip or repo. If not releasing some data, explain public sources and how to regenerate.
6. Experimental details	Yes	Methods	4038 validation samples; 6 LLM labels; BGE/MPNet variants; 40 brain subjects; 48 ROIs; logistic regression; CV details.
7. Statistical significance	Yes	Results and Appendix	Spearman/Pearson, AUC, bootstrap CI, permutation p-values, shuffle controls; define resampling unit.
8. Compute	Yes	Appendix compute table	M4 Pro 24GB macOS 15.3.1 CPU for brain; LLM compute details from collaborator; runtime by component.
9. Code of Ethics	Yes	Ethics / checklist	Public de-identified OpenNeuro data; no new human recruitment; no harmful deployment.
10. Broader impacts	Yes or N/A with expla nation	Limitations / ethics	Foundational interpretability; risk is overinterpretation of brain-LLM similarity; no direct deployment.
11. Safeguards	N/A unless releasing model	Checklist	No new high-risk pretrained model release; analysis code/derived metrics only.
12. Licenses	Yes	Data statement / appendix	OpenNeuro dataset version/license; NeuroEmo citation; model licenses; code package licenses.
13. Assets	Yes if derived assets released	Supplement data card	Document processed embeddings/ROI matrices, provenance, license, limitations.
14. Crowdsourcing/human subjects	N/A for new collection	Ethics statement	No crowdsourcing and no new human subjects. Original dataset creators handled participants.
15. IRB	N/A or explain secondary use	Ethics statement	Public de-identified data only; no new IRB because no new participants/data collection by authors.
16. LLM usage	Yes for analysed LLMs	Methods	LLMs are core objects of study. Writing/editing assistance only if required by final policy; do not list tools as authors.

14. Supplementary material structure

The supplement should make the paper feel complete and reproducible without overloading the nine-page main text. A strong supplement can protect the main paper from reviewer complaints about missing details.

Supplement section	Content to include	Reason
A. Full mathematical definitions	All formulas: centroid, distance, margin, uncertainty, softmax, silhouette, AUC, permutation p, RDM, projection removal.	Supports checklist and clarity.
B. Full LLM phase tables	Complete Phase 1-5 numbers, including per-model variants and outputs.	Keeps main text clean while preserving detail.
C. Brain dataset and preprocessing	Dataset source, BOLD, ROI extraction, 40 subjects, emotion labels, final matrix, scaling.	Human-data transparency.
D. Statistical tests	Exact test settings, bootstrap counts, permutation counts, random seeds, p-value conventions.	Reviewer-proof significance reporting.
E. Additional figures	All PCA maps, density curves, ambiguity gradients, decay curves, VA plots, RDM heatmaps.	Prevents main text figure overload.
F. Reproducibility instructions	Environment, commands, expected output files.	Checklist item 4/5.
G. Ethics and licenses	OpenNeuro license, model licenses, no new participants, data-access notes.	Checklist item 9/12/15.
H. Glossary	BOLD, ROI, RDM, margin, logits, silhouette, AUC, LOSO.	Useful for mixed ML/neuroscience reviewer pool.

15. Data and code package checklist

Folder/file	Required contents	Submission note
README.md	One-command reproduction overview; expected runtime; hardware note; data download instructions.	Must be anonymized.
environment.yml or requirements.txt	Python version and package versions: numpy, scipy, scikit-learn, torch/transformers if used, nilearn for brain.	Avoid unpinned environments if possible.
scripts/01_extract_llm_embeddings.py	Model loading, hidden state extraction, mean pooling, saving embeddings/logits.	Can be a notebook if well documented, but scripts are better.
scripts/02_llm_geometry.py	Centroids, silhouette, overlap, radial bins, certainty buffer.	Regenerates Phase 1-2 tables.
scripts/03_signal_retention.py	Classifier SVD, recursive projection removal, erasure points.	Regenerates Phase 3.
scripts/04_affective_subspace.py	Top-20 projection, silhouette/accuracy, relational centroid distances.	Regenerates Phase 4.
scripts/05_logit_consistency.py	Overlap-logit agreement, logit differences, geometry-logit correlation.	Regenerates Phase 5.
scripts/06_brain_geometry.py	LOSO decoding, brain centroids/RDM, margin-uncertainty, shuffle, VA.	Regenerates brain results.
data/README.md	Source URLs and license notes; do not include restricted raw data unless allowed.	Explains what to download.
results/*.json/*.csv	Reported outputs, including consistency_summary.json and statistical summaries.	Makes verification easy.
figures/	Final figure scripts and generated figures.	Keep figure filenames matching paper references.

16. Decision tree for what belongs in main text vs appendix

Content	Main text?	Appendix?	Reason
Margin formula and interpretation	Yes	Yes with more detail	Core contribution.
Phase 5 logit outputs	Yes	Yes	Strongest LLM mechanism evidence.
Full silhouette ranking all 8 variants	Condensed main figure/table	Full table appendix	Important but not the entire story.
All density curves	One summary or none	Full appendix	Density is supporting and partly negative.
Phase 3 full 0-200 curve	Condensed main plot	Full curve appendix	Main text only needs erasure points and early decay.
20D subspace results	Yes	Full table appendix	Supports low-rank affective core.
Brain LOSO and margin	Yes	Detailed appendix	Essential biological validation.
Brain RDM pairwise table	Optional main small heatmap	Full table appendix	Useful but not central.
VA analysis	Brief main or discussion	Full appendix	Important for global structure but not core proof.
Compute table	Appendix or checklist	Yes	Needed for NeurIPS checklist; not usually main text.
Licenses and ethics details	Brief main/limitations	Full appendix/checklist	Required but not central narrative.

17. Final pre-submission quality gates

Before upload, the paper should pass the following gates. If one gate fails, fix it before formatting polish.

Gate	Question	Pass condition
Claim-evidence gate	Does every abstract claim have a figure/table/statistic?	Yes. If not, weaken the claim or move it to future work.
Metric consistency gate	Is the margin sign convention consistent everywhere?	Yes. One convention, one interpretation.
Statistical gate	Are all main effects accompanied by p-values, CIs, AUCs, or controls?	Yes for central claims.
Reproducibility gate	Can another researcher recreate the main tables from your supplement?	Yes, using README and scripts.
Anonymity gate	Do PDFs, code, notebook paths, and metadata reveal authors/institution?	No. Scrub local paths and names.
Ethics gate	Is public human data use clearly explained?	Yes. OpenNeuro/BIDS, no new participants.
Reviewer confusion gate	Would a reviewer understand why distance fails but margin works?	Yes. Figure and explanation included.
Narrative gate	Does the paper read as principle -> evidence -> validation rather than experiment list?	Yes. If not, reorder Results.

18. Final one-page execution checklist

Before writing	During writing	Before submission
Lock contribution type as General with conceptual/mechanistic contribution.	Keep the narrative order: problem -> distance fails -> margin principle -> LLM proof -> brain validation -> limitations.	Run the claim-evidence audit: every abstract and intro claim maps to a statistic or figure.
Lock margin sign convention and use it consistently in all formulas and figures.	Put Phase 5 geometry-logit coupling at the centre of the LLM results, not in the appendix only.	Check all p-values: never report p=0.0; use p<0.001 or the appropriate permutation bound.
Choose the main figures and move exploratory plots to the appendix.	Write Methods to reproduce the pipeline without opening notebooks.	Verify code/supplement anonymity: no names, local paths, institution references, or non-anonymous links.
Confirm LLM compute details from collaborator.	Use cautious language: coupled/aligned/predictive unless a causal intervention was run.	Include checklist, references, appendices, and PDF in the correct NeurIPS order.
Confirm licenses for OpenNeuro, NeuroEmo, BGE, MPNet, datasets, and released code.	State limitations early enough that reviewers do not feel they discovered them.	Prepare rebuttal-ready answers for distance vs ambiguity, causation, dataset size, novelty, and metric choice.

Final strategic instruction: Every section should answer one question: how does this evidence support the claim that ambiguity is governed by geometric competition? If a paragraph does not answer that question, it probably belongs in the appendix or should be removed.